

Supraventricular Tachycardias

Key

- ⬤ Increased automaticity
- ↻ Re-entry circuit

Sinus Tachycardia

Regular rhythm, normal P waves
Identify/address underlying process

Atrial Tachycardia

Regular, non-sinus P waves
AV nodal blocker

AVNRT

Regular, no sinus P wave
Interrupt re-entry circuit / adenosine

AVRT

Regular, no sinus P wave, "delta"
Interrupt re-entry circuit / adenosine

Atrial Flutter

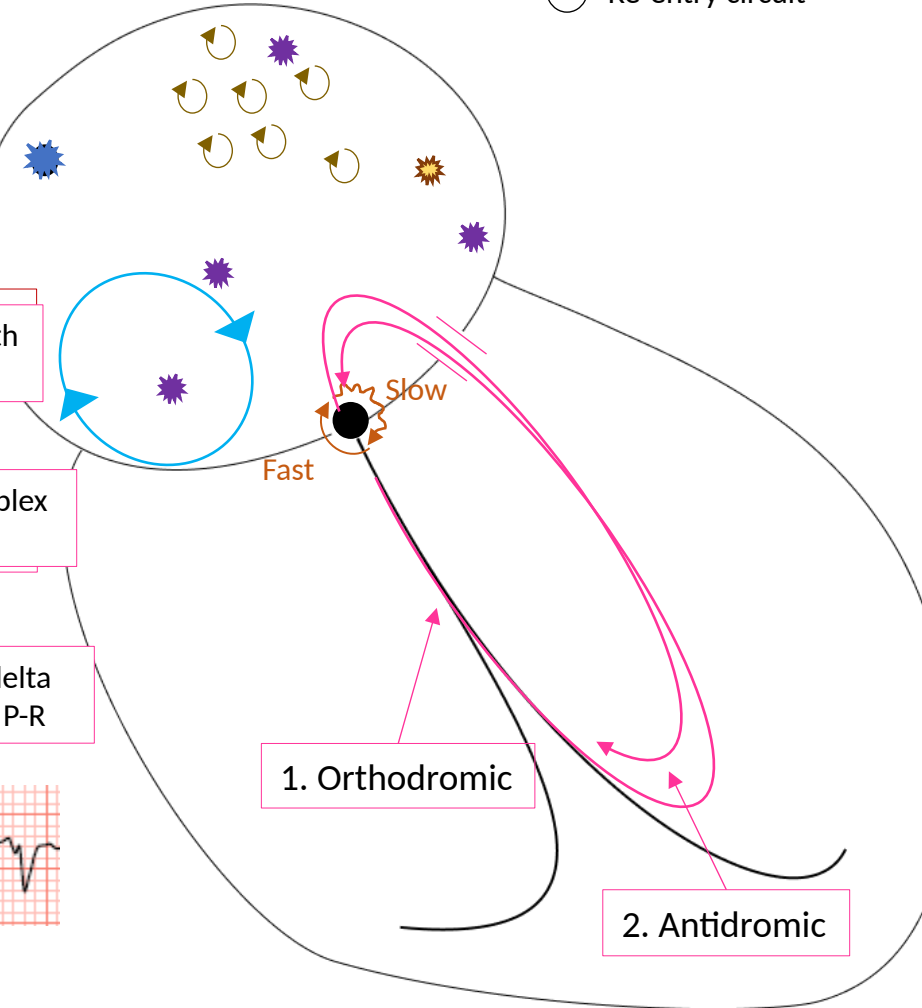
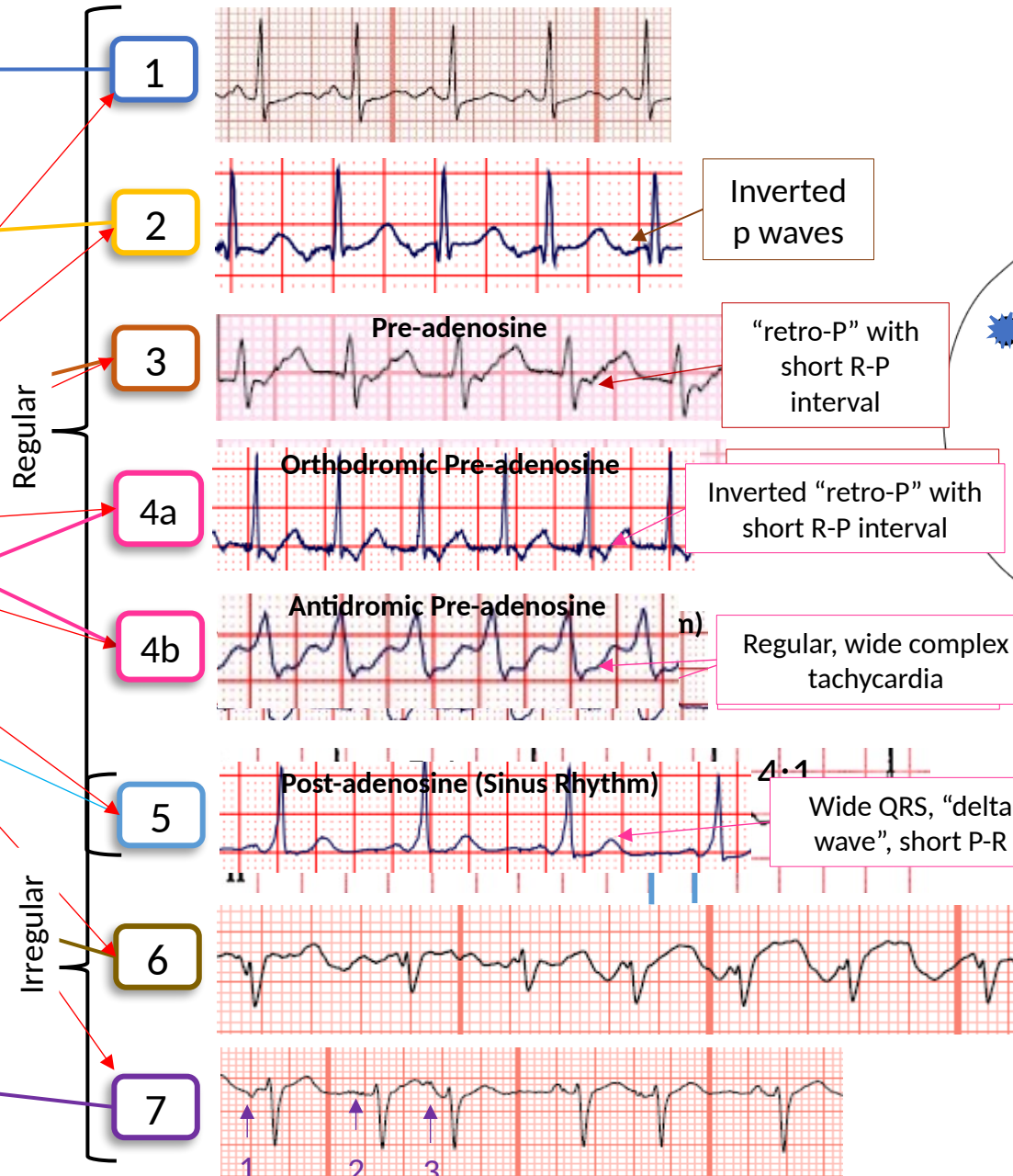
Regular or irregular, "saw tooth" p
Rate control

Atrial Fibrillation

Irregularly irregular, no 'p' waves
Rate control

Multifocal Atrial Tachycardia

Irregular, multiple 'p' morphologies



Acute management — guideline check & safety pearls

Framework unchanged: the 2015 ACC/AHA/HRS and 2019 ESC SVT guidelines keep the same approach taught here — vagal maneuvers first, then adenosine for regular SVT.

Modified (REVERT) Valsalva: semi-recumbent strain into a 10-mL syringe, then immediately lay flat with passive 45° leg raise — converts ~43% vs ~17% for standard Valsalva. Try it before adenosine.

Adenosine is diagnostic AND therapeutic: terminates AVNRT/AVRT, transiently unmasks atrial flutter/atrial tach (reveals the P waves), and does not convert VT — so a response also helps classify the rhythm.

SAFETY — pre-excitation: in pre-excited (WPW) atrial fibrillation — irregular, wide, very fast — AVOID AV-nodal blockers (adenosine, CCB, beta-blocker, digoxin): blocking the node pushes conduction down the accessory pathway and can precipitate VF. Use procainamide or synchronized cardioversion.

Unstable at any point → synchronized cardioversion. Source: Page RL et al. 2015 ACC/AHA/HRS SVT Guideline, Circulation 2016;133:e506; Brugada J et al. 2019 ESC SVT Guideline, Eur Heart J 2020;41:655.

Take-Home Points

- All tachycardias arise from either increased automaticity (nodal/ectopic pacemaker) or a re-entry circuit — re-entry circuits are generally faster.
- There are only 7 SVTs; sort them by two features — regular vs. irregular rhythm and the presence/morphology of P waves.
- Stable SVT is managed three ways: address the underlying process (sinus tach, atrial tach, MAT), interrupt the re-entry circuit with vagal maneuvers/adenosine (AVNRT, AVRT), or rate-control with AV-nodal blockers (AF, atrial flutter).
- Adenosine reveals the underlying rhythm: it resolves AVNRT/AVRT, unmasks atrial flutter/tach, and does not affect VT.
- Extreme right-axis deviation (positive in aVR) is a quick, reliable clue for VT — keep it in mind whenever a regular wide-complex tachycardia could be antidromic AVRT vs. VT. Avoid AV-nodal blockers in pre-excited AF.

Attribution & Optimization Notes

Adapted from TeachIM.org — "Supraventricular Tachycardia (SVT)" (Brandon Fainstad, MD; expert review Zach Goldberger, MD).

Files: presenter board SVT-interactive-board-for-presenting-1.pptx; companions SVT-interactive-board-for-learning.pptx, SVT-Practice-ECGs.pptx, SVT-learner-handout-1.pdf, SVTs-handout-1.pdf.

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Optimization (June 2026): guideline framework verified current (2015 ACC/AHA/HRS, 2019 ESC). Surgical corrections: "Multi-Atrial Tachycardia" → "Multifocal Atrial Tachycardia (MAT)" (standard term for the MAT abbreviation) on the presenter AND learning boards; "Orhtodromic" → "Orthodromic" typo on the presenter board. All graphics and click-to-reveal animations preserved.

Added: acute-management currency/safety slide (modified REVERT Valsalva; pre-excited-AF AV-nodal-blocker caution), this take-home slide, and attribution. Companion package: HTML launcher, one-pager PDF, Anki, ABIM MCQs, dotphrase. Clinician-review before teaching.